

Jobverse

Project Roadmap

A job discovery and application automation system, built on n8n, Google Sheets, RapidAPI (JSearch), Groq AI, and a React operator dashboard.

1. System Overview

Piece	What it does	Built on
Workflow 1 — Job Discovery	Searches for jobs per candidate, AI-scores them against the candidate's profile, saves qualified matches	n8n + RapidAPI (JSearch) + Groq
Candidate Intake & Scheduler	Collects candidate profiles via a Google Form, runs the daily search loop across all active candidates	n8n + Google Forms/Sheets
Workflow 2 — Application Preparation (not yet built)	Tailors a CV/cover letter to each shortlisted job, drafts the email application for human review	n8n + Groq + Gmail
Jobverse Dashboard	Operator-facing UI to manage candidates, review sourced jobs, and (eventually) applications	React + a small custom n8n-based API

All persistent data lives in one Google Sheet with multiple tabs: Candidates, Shortlisted Jobs, Sessions, and eventually Applications.

2. Current Status

✓ Built and debugged (working)

- Job Discovery pipeline: RapidAPI/JSearch integration, response flattening/normalization, job validation, stable job-ID generation, recency filtering, deduplication.
- AI scoring: Groq-based multi-factor match scoring, with a bounded retry (different model on retry) and rate-limit throttling via a Split-In-Batches + Wait loop.
- Multi-candidate architecture: split into a reusable “Job Discovery – Per Candidate” sub-workflow, called either immediately (new form signup) or in a daily loop (scheduled, one candidate at a time) by the Candidate Intake & Scheduler parent workflow.
- Google Sheets persistence: per-candidate scoped deduplication and shortlist saving (matched on jobId + candidateEmail, so candidates' results never bleed into each other).
- Jobverse frontend (UI only): React dashboard with working sidebar navigation across all 9 planned pages; Candidates and Prospects pages are fully interactive (edit target, toggle active, delete, filter by candidate).
- Jobverse API layer: a session-token auth system (login → token → validated per request) and five webhook endpoints (login, list/update/delete candidates, list prospects, manual generate trigger), built and validated for structural correctness.
- Frontend deployed: live at <https://jobverse-4d3q.onrender.com>.

□ Built, not yet verified end-to-end

- The Jobverse API workflow has not yet been confirmed working against a live n8n instance (curl testing was the next step, not yet confirmed complete).
- CORS is currently open (*); tightening to the real frontend domain via an environment variable was planned but not yet confirmed done.

□ Designed but not built

- Workflow 2 — Application Preparation: channel scope (email-only), CV handling (upload + extract), and autonomy default (draft-and-hold via Gmail draft) were all decided, but no n8n nodes exist yet.
- CV/cover-letter tailoring by type (NHS, corporate, technical, executive, and standard): raised as a requirement but never resolved — specifically, whether NHS applications need a structurally different format (person-specification/competency answers) rather than just a different tone of the same CV.

○ not started

- Wiring the live frontend to the real API (replacing sample data with fetch() calls + token handling).
- The 7 remaining dashboard pages: Review Queue, Applications, Generate, Follow-Ups, Reports, Activity, Help & Guide.
- Response/reply tracking (needed before Follow-Ups can function).

3. Roadmap — In Dependency Order

Phase 1 — Prove the API layer works

- ☐ Import Jobverse_Auth_Check.json and Jobverse_API.json into n8n.
- ☐ Add the Sessions tab and the two missing Candidates columns (applicationsSentCount, applicationTarget).
- ☐ Set JOBVERSE_OPERATOR_PASSWORD on the n8n server.
- ☐ Wire up Google Sheets credentials and the two sub-workflow references.
- ☐ Activate the workflow, confirm login + list-candidates via curl.

Nothing downstream is worth doing until this is confirmed working.

Phase 2 — Connect the deployed frontend to real data

- ☐ Replace sample-data arrays in the React app with real fetch() calls.
- ☐ Handle login (store token in memory) and attach it as a Bearer header on every request.
- ☐ Re-deploy to jobverse-4d3q.onrender.com.

Phase 3 — Lock down security now that a real URL exists

- ☐ Set JOBVERSE_FRONTEND_URL env var to https://jobverse-4d3q.onrender.com.
- ☐ Update the 6 Respond nodes' CORS header from * to the env-var expression.
- ☐ Re-test with an Origin header to confirm.

Phase 4 — Resolve the open CV-tailoring question, then build Workflow 2

- ☐ Decide: are NHS/corporate/technical/executive/standard purely tone-and-formatting variants of one AI prompt, or does NHS require a structurally different output?
- ☐ Extend the Google Form with a CV upload question.
- ☐ Add Download CV File (Google Drive) → Extract From File to the intake pipeline; store extracted text on the candidate record.
- ☐ Build the email-detection, AI drafting, and Gmail-draft-creation nodes.
- ☐ Add the Applications sheet tab and version-control logging.

Phase 5 — Build the dependent frontend pages

- ☐ Review Queue (from Applications, status = Pending Review)
- ☐ Applications (full log)